SP_N and SDP_N Verification and Validation

Brantley–Larsen and C5G7 against Carreño et al. (2024)

GARNET V&V report, audit item O10

30 October 2026

Summary

This report verifies GARNET's SP_N and SDP_N solvers against the two two-dimensional problems of Carreño et al. (2024), the paper whose equations they implement, after the October 2026 code audit (item O10).

- **Brantley–Larsen, code verification.** With the paper's coupled Marshak vacuum, GARNET's mesh limits equal the paper's Table 6 within 0.6 pcm for SP1, SP3, SP5, SP7, SDP1 and SDP2. Convergence is second order (observed 1.93–2.00), and the extrapolation is good to about 0.2 pcm.
- **SDP3 coefficient tables.** The paper's SDP3 results were computed with the tables in the authors' FEMFFUSION code: with them, GARNET reproduces Table 6 within 0.2 pcm. Those tables differ from the paper's printed Eq. (A.6) (which gives +24 pcm) and from the first-principles tables GARNET uses (−10 pcm on Brantley–Larsen, about −14 pcm on C5G7).
- **C5G7, pin-resolved.** The pin powers reproduce the paper's Table 7 AVG and RMSE against the NEA MCNP reference within 0.04 percentage points (AVG 1.43% for diffusion down to 0.53–0.57% for SP5, SP7 and SDP2). The eigenvalues are 11–22 pcm below the paper's (42 pcm for SDP3 with GARNET's tables), a model difference of 3–5% of the angular error whose source is not identified. Both codes show the paper's C5G7 conclusion: k stagnates with angular order while the pin powers improve.
- **Flux shapes.** Along the lines the paper plots, GARNET agrees with the OpenMOC and OpenMC profiles as closely as the paper's own solutions do (0.4% of the peak on Brantley–Larsen, 1.1–1.7% on C5G7 for SDP3).
- **Findings for the code** (Section 6): the per-moment Robin vacuum, then the default of every solver except SP3, is 52–73 pcm low on a leaky core, so Marshak is now the default wherever it is available, with Robin kept for comparison; the C5G7 builder's sampled pin areas misstate the fuel inventory and move k by up to 226 pcm (exact areas are now an option); and the non-symmetric SDP_N solve failed to converge at 2.7 million cells.
- **Tests.** The Brantley–Larsen mesh-limit test now covers every method at 4 pcm (was SP3 and SDP1 at 10 pcm); a slow pin-resolved C5G7 test checks diffusion against the paper's k and pin powers.

1 Scope and references

The reference is A. Carreño, A. Vidal-Ferràndiz, D. Ginestar and G. Verdú, "Development of a diffusive simplified double P_N approximation to the neutron transport equation", *Ann. Nucl. Energy* 207 (2024) 110675 (CC BY 4.0, in `literature/`). GARNET's SP_N ($N = 1, 3, 5, 7$) and SDP_N ($N = 1, 2, 3$) solvers implement the same equations in the same unified form. The paper solves them with second-degree finite elements; GARNET uses cell-centred finite volumes. Two of the paper's problems are used:

- **Brantley–Larsen** (paper Sec. 4.2): one group, three fuel bars in a 10×10 cm moderator square, reflective at $x = 0$ and $y = 0$, vacuum at $x = 10$ and $y = 10$ (paper Fig. 3, Table 5). References:

the paper's own SP_N/SDP_N eigenvalues (Table 6), OpenMOC ($k = 0.80536$) and TWODANT S_{16} ($k = 0.80613$), and the OpenMOC flux along $y = 4.5$ cm (Fig. 4).

- **C5G7** (paper Sec. 4.3): the OECD/NEA two-dimensional quarter core, seven groups, vacuum outer boundary (Fig. 5). References: the paper's eigenvalues and pin-power errors (Table 7), MCNP ($k = 1.18655$), the NEA MCNP pin powers, and the OpenMC flux along two assembly diagonals (Figs. 6–7).

Three questions are separated. *Solution verification*: how far each `GARNET` eigenvalue is from its own mesh limit. *Code verification*: whether the mesh-converged `GARNET` result equals the paper's solution of the same equations. *Validation of the approximation*: how far SP_N and SDP_N are from transport. The last is a property of the equations, so `GARNET` should reproduce the paper's distance from transport rather than improve on it.

2 Code, models and metrics

Code state. Commit `20967d3` (the merged 2026-10-28 audit) plus the uncommitted audit work of 29–30 October. The audit changes that bear on these problems are: the coupled Marshak vacuum is now the default of the dedicated SP3 solver; plain power iteration stops on an error bound rather than the last change; non-symmetric SDP_N blocks default to GMRES; and, added for this study, `build_c5g7_2d` accepts exact pin areas (`exact_area=True`) and a vacuum outer boundary (`outer_bc="vacuum"`). Both new options are off by default, so no earlier result moves.

Discretization. Cell-centred finite volumes on a uniform Cartesian mesh, second order where material interfaces lie on cell faces. In the pin-resolved C5G7 model a cell cut by the $r = 0.54$ cm fuel circle is a volume blend of pin and water: linear in the reaction cross sections, harmonic in the diffusion coefficients, and weighted by fission production in the emission spectrum. Its covered fraction is computed in closed form. The pin-cell-homogenized model, which `tests/validation/test_c5g7.py` runs, volume-weights each 1.26 cm pin cell instead.

Vacuum boundaries. *Marshak* is the coupled condition $-\mathbf{n} \cdot D\nabla U = (g \otimes I)\, U$ that the paper uses. *Robin* applies $\alpha = 1/2$ to each moment separately. SP1 is diffusion, for which the two coincide. When these runs were made only the dedicated SP3 solver defaulted to Marshak; following this study (finding 3), every SP_N/SDP_N eigen, transient and noise solver now does so wherever the coupled term exists, and Robin stays available as `marshak_vacuum=False`. The dedicated SDP1 solver has no Marshak term and keeps Robin; its Marshak runs use the general SDP_N block of order 1. Every run here names its boundary explicitly, so the change of default moves none of the results.

Tolerances. Brantley–Larsen: `tol_k` $= 10^{-10}$, `tol_source` $= 10^{-9}$. C5G7: 10^{-9} and 10^{-8}. Every run converged.

Metrics. Eigenvalue differences are signed, $10^5(k - k_{\mathrm{ref}})$ in pcm. The paper's Eq. (49) defines a relative difference, but its Tables 6 and 7 list $10^5|k - k_{\mathrm{ref}}|$ (for example SP1 in Table 7: $10^5 \times |1.18336 - 1.18655| = 319$); the tables are what is compared here. Pin-power errors $e_i = (P_i - P_i^{\mathrm{ref}})/P_i^{\mathrm{ref}}$ over the 1056 fuel pins, with powers normalized to a mean of 1: AVG $= \langle |e_i| \rangle$ (paper Eq. 50); the paper's RMSE, whose Table 7 values match $\sqrt{\sum e_i^2}/N$ (Eq. 51 as printed equals AVG); the conventional RMS $\sqrt{\langle e_i^2 \rangle}$; MRE $= \sum |e_i| P_i^{\mathrm{ref}} / \sum P_i^{\mathrm{ref}}$; and the largest $|e_i|$. Flux profiles are compared after a least-squares scale factor (the paper's flux units are arbitrary), as RMS and largest deviation in percent of the reference peak.

Paper figures. The paper's plots are vector graphics, so the reference curves were read from the drawn paths themselves (`extract_paper_figures.py`), not digitized from pixels. Axis calibration is a linear fit through the tick marks; the residual plus half the line width is below 0.4% of the curve's peak in all five panels.

3 Brantley–Larsen

3.1 Problem and runs

One energy group: fuel $\Sigma_t = 1.5$, $\Sigma_{s0} = 1.35$, $\nu\Sigma_f = 0.24$ cm^{-1}; moderator $\Sigma_t = 1.0$, $\Sigma_{s0} = 0.93$ cm^{-1} (paper Table 5). The fuel bars occupy $x \in [1,2] \cup [4,5] \cup [7,8]$, $y \in [0,9]$ cm. Every interface lies on a cell face for n a multiple of 10, so the finite-volume scheme is second order. Each method was run with Marshak and Robin vacuum on $n \times n$ meshes, $n = 40, 80, 160, 320, 640$. The GMRES runs of SDP2 (Marshak) and SDP3 (both boundaries) at $n = 640$ would have taken 8–20 h each on the shared node and were stopped; for those three the extrapolation uses $n = 80, 160, 320$.

3.2 Mesh convergence

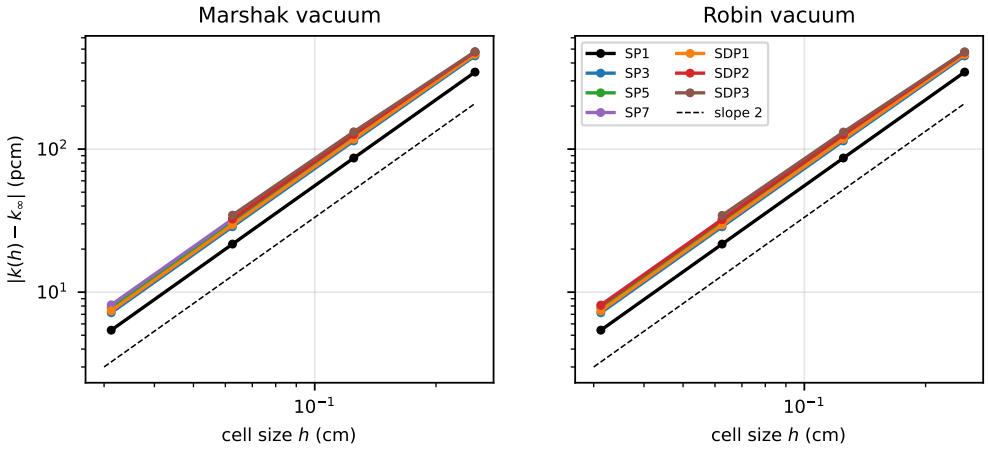


Figure 1: Brantley–Larsen: distance of each mesh's k from its own extrapolated limit. Every method and boundary treatment converges at second order.

Table 1 and Fig. 1. The observed order from the finest three meshes is between 1.93 and 2.00 for every method and both vacuum treatments, and k decreases monotonically under refinement. k_∞ is the Richardson extrapolation with the observed order. The GCI column is Roache's grid-convergence index with safety factor 1.25; it is 1.7–2.6 pcm where $n = 640$ exists and 10–11 pcm where the triple ends at $n = 320$. The GCI is a conservative bound: wherever both triples exist (14 configurations), extrapolating from $n = 80, 160, 320$ and from $n = 160, 320, 640$ agrees within 0.21 pcm.

Table 1: Brantley–Larsen eigenvalues by mesh, observed order p, extrapolated k_∞ and GCI.

Method	Vacuum	k_{80}	k_{160}	k_{320}	k_{640}	p	k_∞	GCI (pcm)
SP1	–	0.777667	0.777018	0.776855	0.776814	2.00	0.776801	1.70
SP3	Marshak	0.800178	0.799323	0.799109	0.799055	2.00	0.799037	2.25
	Robin	0.799648	0.798795	0.798580	0.798526	2.00	0.798508	2.25
SP5	Marshak	0.804023	0.803112	0.802881	0.802823	1.99	0.802803	2.43
	Robin	0.803488	0.802578	0.802347	0.802289	1.99	0.802270	2.43
SP7	Marshak	0.804814	0.803869	0.803627	0.803566	1.99	0.803546	2.57
	Robin	0.804273	0.803331	0.803089	0.803028	1.99	0.803007	2.56
SDP1	Marshak	0.802787	0.801908	0.801687	0.801631	2.00	0.801613	2.32
	Robin	0.802056	0.801178	0.800957	0.800901	2.00	0.800883	2.32
SDP2	Marshak	0.804992	0.804050	0.803809	–	1.97	0.803726	10.36
	Robin	0.804464	0.803524	0.803284	0.803223	1.99	0.803203	2.54
SDP3	Marshak	0.805241	0.804266	0.804011	–	1.93	0.803921	11.30
	Robin	0.804699	0.803728	0.803474	–	1.93	0.803383	11.26

On the 80×80 mesh of `tests/validation/test_brantley_larsen_2d.py` the mesh error is +87 pcm for SP1 and +114 to +132 pcm for the transport methods (Table 3). That is larger than the difference between consecutive SP_N orders above SP5, so single-mesh comparisons with the paper on that mesh mostly measure mesh error.

3.3 Comparison with the paper

Table 2: Brantley–Larsen: extrapolated `GARNET` eigenvalues against the paper's Table 6 and OpenMOC (pcm). "Robin − paper" uses `GARNET`'s Robin-vacuum k_∞.

Method	Paper k	Paper − OpenMOC	`GARNET` k_∞	− paper	Robin − paper	`GARNET` − OpenMOC
SP1	0.77680	-2856	0.77680	+0.1	–	-2856
SP3	0.79904	-632	0.79904	-0.3	-53.2	-632
SP5	0.80280	-256	0.80280	+0.3	-53.0	-256
SP7	0.80354	-182	0.80355	+0.6	-53.3	-181
SDP1	0.80161	-375	0.80161	+0.3	-72.7	-375
SDP2	0.80373	-163	0.80373	-0.4	-52.7	-163
SDP3	0.80402	-134	0.80392	-9.9	-63.7	-144

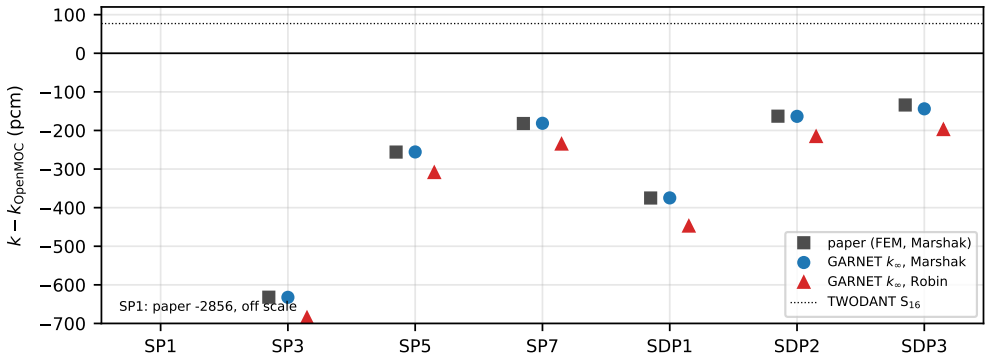


Figure 2: Brantley–Larsen eigenvalues relative to OpenMOC: paper (finite elements, Marshak), `GARNET` mesh limit with Marshak and with Robin vacuum. TWODANT S_{16} is the dotted line.

With the Marshak boundary, `GARNET`'s mesh limit equals the paper's Table 6 within 0.6 pcm for SP1, SP3, SP5, SP7, SDP1 and SDP2 (Table 2, Fig. 2), about the ±0.5 pcm rounding of Table 6. This is the code verification: the same equations, discretized independently (finite volumes here, second-degree finite elements in the paper), give the same eigenvalue.

SDP3 is −9.9 pcm from the paper. `GARNET` does not use the SDP3 coefficient rows 2–3 printed in the paper's Eq. (A.6): a from-scratch symbolic derivation from the paper's own closure reproduces its SDP1 and SDP2 tables and every Marshak matrix exactly, but not those rows (`tests/verification/test_sdpn_derivation.py`, note in `garnet/spn.py`). Section 3.7 shows that the paper's SDP3 values come from a third set, the tables in the authors' FEMFFUSION code, and that with those `GARNET` reproduces them too.

The Robin vacuum lowers k by 52–54 pcm for every method except SDP1 (−73 pcm). It was the default of every solver except the dedicated SP3 when this study started; Marshak is now the default wherever it is available, and Robin is kept for comparison.

The angular error, $k_\infty - k_{\mathrm{OpenMOC}}$ with Marshak, is −2856 pcm for diffusion, −632, −256 and −181 pcm for SP3, SP5, SP7, and −375, −163 and −144 pcm for SDP1, SDP2, SDP3. At equal numbers of moments SDP_N is closer to transport than SP_{2N+1} (SDP1/SP3, SDP2/SP5, SDP3/SP7), the paper's central claim, and it holds with the re-derived SDP3 tables. The two transport references differ by 77 pcm (TWODANT

S_{16} above OpenMOC), which is about half the remaining SDP3 and SP7 errors: the references do not resolve which of those two is closer to transport.

3.4 Error budget on the test mesh

Table 3: Brantley–Larsen error budget (pcm). Mesh error is $k_{80} - k_\infty$ with Marshak. The last two columns are the single-mesh comparison of the validation test on the 80×80 mesh, with each vacuum treatment.

Method	Mesh error at $n = 80$	Robin − Marshak	Angular ($k_\infty - k_{\text{OpenMOC}}$)	k_{80}− paper, Marshak	k_{80}− paper, Robin
SP1	+86.6	–	-2856	+86.7	–
SP3	+114.1	-52.9	-632	+113.8	+60.8
SP5	+121.9	-53.3	-256	+122.3	+68.8
SP7	+126.8	-53.8	-181	+127.4	+73.3
SDP1	+117.4	-73.0	-375	+117.7	+44.6
SDP2	+126.6	-52.3	-163	+126.2	+73.4
SDP3	+132.0	-53.7	-144	+122.1	+67.9

Table 3 splits the single-mesh comparison into its parts. With Marshak, every method is +87 to +127 pcm above the paper on the 80×80 mesh, and all of it is mesh error (for SDP3, less the −10 pcm of its re-derived tables). With Robin the offset is smaller, +45 to +73 pcm, only because two errors of opposite sign add: mesh error and the Robin offset (−52 to −73 pcm). Under the old Robin defaults the test's 150 pcm envelope therefore passed partly by cancellation; it now checks the Marshak results, and the Robin offset separately.

3.5 Equivalent code paths

Table 4: Dedicated solvers against the general SP_N/SDP_N block of the same order.

Comparison	Vacuum	max $\lvert\Delta k\rvert$, $n = 40$–640	Note
SP3 dedicated vs SPN block	Marshak	2.2e-10	same condition
SP3 dedicated vs SPN block	Robin	6.8e-06	per-moment term on different moments
SDP1 dedicated vs SDPN block	Robin	3.5e-06	per-moment term on different moments

SP3 and SDP1 each have a dedicated two-moment solver and can also be run through the general block of order 1 (Table 4). With Marshak, the dedicated SP3 solver and the SP_N block agree to 2×10^{-10} on every mesh: two independent implementations of the coupled boundary. With Robin they differ by up to 0.7 pcm (SP3) and 0.35 pcm (SDP1), because the dedicated solvers apply $\alpha = 1/2$ to (Φ_1, ϕ_2) and the general block to its U moments. The per-moment Robin term is an approximation whose result depends on the choice of moments; the Marshak condition is not.

3.6 Flux along $y = 4.5$ cm

Fig. 3 and Table 5. `GARNET`'s SDP3 and SP7 deviate from OpenMOC by 0.36–0.37% of the peak (RMS), the paper's own SDP3 and SP7 markers by 0.39–0.40%. Both are at the extraction uncertainty (0.39%), so on this line the two codes agree with transport equally well. The remaining structure, up to 1.8% of the peak, sits at the fuel-bar edges, where the angular flux is most anisotropic; SP3 is the largest there and SDP3 the smallest. The paper's markers, 0.2 cm apart, are too sparse to show it.

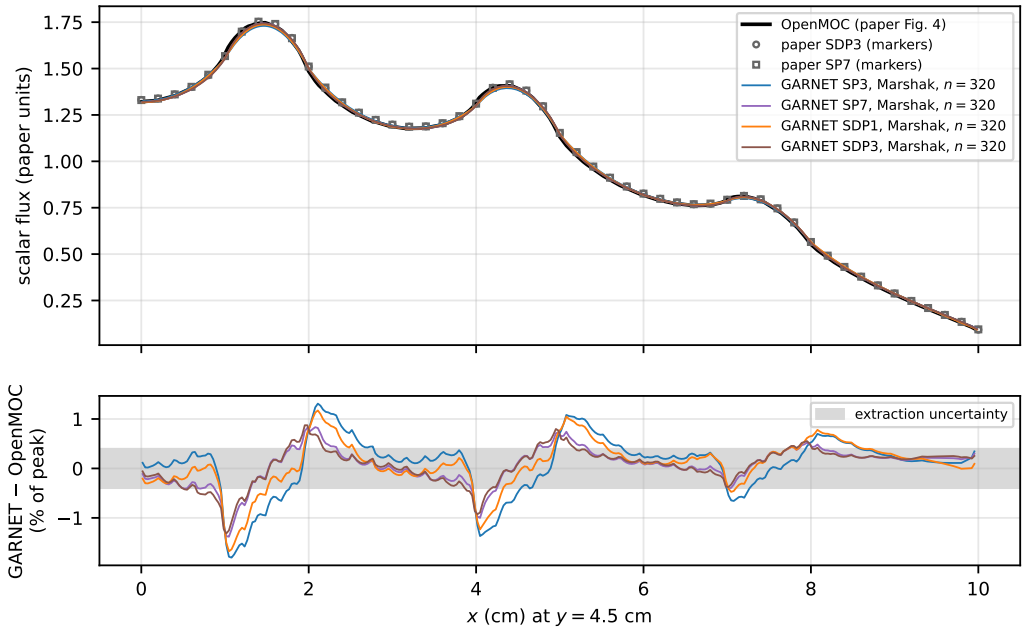


Figure 3: Scalar flux along $y = 4.5$ cm: OpenMOC and the paper's SDP3 and SP7 markers (paper Fig. 4) and `GARNET` with Marshak vacuum. `GARNET` curves are scaled to OpenMOC by least squares. Lower panel: `GARNET` minus OpenMOC; the grey band is the extraction uncertainty.

Table 5: Flux along $y = 4.5$ cm against OpenMOC, after a least-squares scale (% of the OpenMOC peak).

Solution	RMS (% of peak)	max (% of peak)
paper SDP3	0.39	1.40
paper SP7	0.40	1.39
SP3	0.59	1.80
SP7	0.37	1.38
SDP1	0.49	1.68
SDP3	0.36	1.32

3.7 The paper's printed SDP3 tables

Three sets of SDP3 coefficient tables are in circulation, and they differ only in the closure rows 2–3:

- **Re-derived** (`GARNET`'s default): rows 2–3 obtained symbolically from the paper's own closure, Eq. (34). The same derivation reproduces the paper's SDP1 and SDP2 tables and all three Marshak matrices exactly.
- **FEMFFUSION**: the tables in the authors' code (`src/matrix_operators/matrix_operators_sdpn.cc`).
- **Paper (A.6)**: the tables as printed. They equal FEMFFUSION's except row 2 of $c^{(3)}$, printed as $(36/55,\ 54/175)$ where the code has $(9/25,\ -54/175)$.

On Brantley–Larsen (Table 6) the FEMFFUSION tables give a mesh limit 0.2 pcm from the paper's 0.80402, the re-derived tables −9.9 pcm and the printed tables +23.6 pcm. The paper's SDP3 results were therefore computed with the FEMFFUSION tables, and with them `GARNET` reproduces Table 6 for all seven methods. On C5G7 at $s = 4$ the FEMFFUSION tables raise SDP3 by 14.3 pcm and the printed ones by 15.2 pcm.

Table 6: Brantley–Larsen SDP3 with Marshak vacuum and the three coefficient sets. k_∞ is extrapolated from $n = 80, 160, 320$.

SDP3 tables	k_{80}	k_{160}	k_{320}	p	k_∞ (80–320)	− paper
re-derived (`GARNET`)	0.805241	0.804266	0.804011	1.93	0.803921	-9.9
FEMFFUSION code	0.805352	0.804369	0.804111	1.93	0.804018	-0.2
paper Eq. (A.6) as printed	0.805598	0.804609	0.804349	1.93	0.804256	+23.6

Which tables are right is a question of derivation, which this benchmark cannot settle. The re-derived rows are the only ones of the three that follow from the closure the paper states, and `tests/verification/test_sdpn_derivation.py` keeps them in place. The choice moves k by 10–15 pcm on these problems, small against SDP3's angular error (−144 pcm against OpenMOC on Brantley–Larsen).

3.8 Cost

Table 7: Brantley–Larsen cost at $n = 160$, best of three solves on one CPU thread (merlin7 `cn000`). Inner iterations are operator applies of the within-group Krylov solver. The paper's CPU times are for its own 80×80 finite-element mesh and are listed only for scale.

Method	Vacuum	Krylov	outers	inners	solve (s)	paper CPU (s)
SP1	–	CG	18	3357	0.77	1.1
SP3	Marshak	CG	17	3121	1.75	1.5
SP3	Robin	CG	18	3224	1.52	1.5
SP5	Marshak	CG	18	3139	2.66	2.0
SP5	Robin	CG	18	3144	2.37	2.0
SP7	Marshak	CG	18	3088	3.79	2.6
SP7	Robin	CG	18	3094	3.39	2.6
SDP1	Marshak	GMRES	24	4877	8.03	1.5
SDP1	Robin	CG	19	3313	1.63	1.5
SDP2	Marshak	GMRES	26	4738	12.95	2.0
SDP2	Robin	CG	18	3081	2.25	2.0
SDP3	Marshak	GMRES	29	4923	18.71	2.6
SDP3	Robin	GMRES	28	4803	17.58	2.6

Table 7. At $n = 160$ (25 600 cells, one group) the SP_N solves take 0.8–3.8 s with nearly the same inner work for every order (3088–3357 operator applies): the cost grows with the size of the block, not with the number of iterations, and the Marshak boundary costs SP_N nothing measurable (CG either way). SDP1 and SDP2 with Robin vacuum are symmetrized and solved by CG at about the cost of SP3 and SP5. With Marshak their block is non-symmetric and runs GMRES: 1.5 times the applies but 5–6 times the time (8.0 and 13.0 s), the rest being GMRES's orthogonalization. SDP3 has no symmetric form once a vacuum face is present and runs GMRES with either boundary (18–19 s). The paper's CPU times refer to its own mesh and machine and are listed for scale only.

4 C5G7

4.1 Models and runs

The quarter core is 51×51 pin cells of 1.26 cm (four 17×17 assemblies and a 21.42 cm water reflector), seven groups, reflective on the two symmetry planes and vacuum outside. Three `GARNET` models were run:

- **Pin-resolved** (the paper's geometry): every lattice position is an $r = 0.54$ cm cylinder in water, on $s \times s$ cells per pin cell, with exact covered fractions (`build_c5g7_2d(pin_resolved=True, exact_area=True, outer_bc="vacuum")`). With Marshak vacuum, $s = 4$–48 for SP_N and $s = 4$–24 for SDP_N; Robin vacuum at $s = 4, 8, 16$; zero-flux outer faces at $s = 8$.
- **Pin-resolved with sampled areas**: the builder's default 16×16-point sampling of the covered fractions, SP1 only.
- **Pin-cell homogenized**: each pin cell volume-weighted, zero-flux outer faces, $s = 1, 2, 4, 8$. This is the model of `tests/validation/test_c5g7.py` (at $s = 1$).

Up to $s = 12$ runs were made on the CPU, from $s = 8$ on a GH200 (`data/runs-gh200/`). The 18 configurations run on both agree to 1.3×10^{-13} in k.

4.2 Mesh convergence

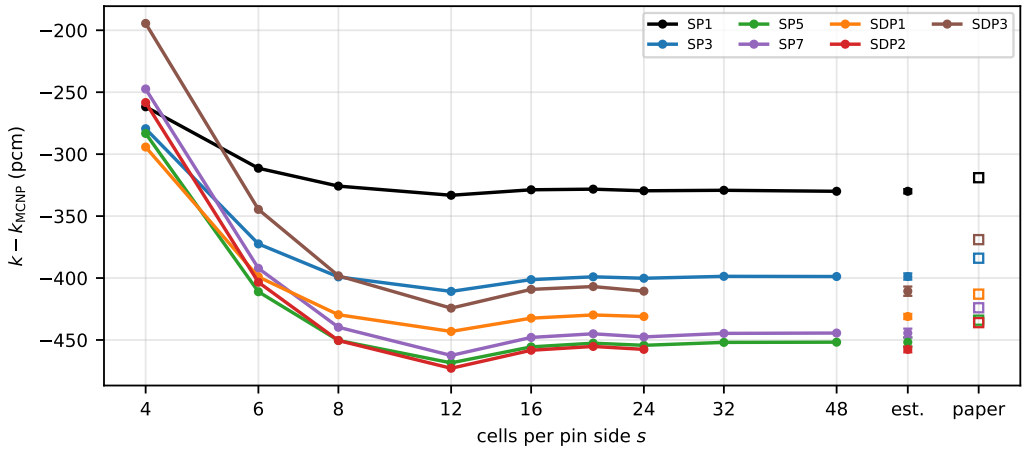


Figure 4: C5G7 pin-resolved, Marshak vacuum: $k - k_{\mathrm{MCNP}}$ against cells per pin side. "est." is the finest-mesh estimate with its spread over $s \geq 16$; "paper" is Table 7.

The pin-resolved model does not converge monotonically (Fig. 4, Table 8). At $s = 4$ every method is 68–216 pcm above its fine-mesh value; at $s = 12$ every method dips 3–18 pcm below it; from $s = 16$ on the values settle. How the fuel circle cuts the cells changes from mesh to mesh, so the blend error does not follow a power law, and Richardson extrapolation through these points would be arbitrary. The estimate is therefore the finest-mesh k ($s = 48$ for SP_N, 24 for SDP_N). Its uncertainty, 1.7–3.8 pcm, is the larger of two measures: the largest difference from it among the meshes with $s \geq 16$, and its distance to a least-squares $k_\infty + b/s^2$ fit over the same meshes.

Table 8: C5G7 pin-resolved eigenvalues, Marshak vacuum. Paper column: Table 7 k and $10^5(k - k_{\mathrm{MCNP}})$. Differences in pcm.

Method	Paper (pcm)	$s = 4$	$s = 8$	$s = 12$	$s = 16$	$s = 24$	$s = 32$	$s = 48$	k estimate	± (pcm)	− MCNP	− paper
SP1	1.18336 (-319)	1.18393	1.18329	1.18322	1.18326	1.18325	1.18326	1.18325	1.18325	1.7	-330	-11
SP3	1.18271 (-384)	1.18376	1.18256	1.18244	1.18254	1.18255	1.18256	1.18256	1.18256	2.5	-399	-15
SP5	1.18221 (-434)	1.18372	1.18205	1.18187	1.18199	1.18201	1.18203	1.18203	1.18203	3.8	-452	-18
SP7	1.18231 (-424)	1.18408	1.18215	1.18193	1.18207	1.18207	1.18210	1.18211	1.18211	3.5	-444	-20
SDP1	1.18242 (-413)	1.18361	1.18225	1.18212	1.18223	1.18224	–	–	1.18224	2.1	-431	-18
SDP2	1.18219 (-436)	1.18397	1.18205	1.18182	1.18197	1.18197	–	–	1.18197	2.4	-458	-22
SDP3	1.18286 (-369)	1.18461	1.18257	1.18231	1.18246	1.18244	–	–	1.18244	3.7	-411	-42

4.3 Comparison with the paper and MCNP

For the six methods whose tables the paper and `GARNET` share (SP1–SP7, SDP1, SDP2), `GARNET`'s estimate is 11–22 pcm below Table 7, and the offset grows with the number of moments: 11 pcm for diffusion, 15–20 pcm for SP3–SP7, 18–22 pcm for SDP1–SDP2. For SDP3 it is 42 pcm. The FEMFFUSION tables raise C5G7 SDP3 by 14 pcm at $s = 4$ (Section 3.7), so with them the offset would be about 27 pcm, in line with the other methods.

Unlike Brantley–Larsen, the two C5G7 models are not the same discrete problem. The paper meshes each pin with second-degree finite elements (its Fig. 5) and states neither how the circular pin boundary is represented nor a mesh study; `GARNET` blends the cells the circle cuts, with exact areas. The source of the 11–22 pcm offset is not identified. It is 3–5% of the 320–460 pcm angular error and changes none of the orderings below.

Against MCNP both codes give the same picture. Diffusion is closest (-330 pcm here, -319 pcm in the paper), and every SP_N and SDP_N order is further away (-399 to -458 pcm here, -369 to -436 pcm in the paper). This is the stagnation the paper reports for C5G7: on this lightly heterogeneous lattice, the eigenvalue is not a measure of angular accuracy. The pin powers are.

4.4 Modelling choices

Table 9: C5G7 at $s = 8$: effect of the vacuum treatment and of pin-cell homogenization on k (pcm).

Method	Robin − Marshak	zero flux − Marshak	homogenized − pin-resolved (zero flux)
SP1	–	-0.23	+379
SP3	-0.20	-1.33	+591
SP5	-0.28	-1.35	+654
SP7	-0.29	-1.36	+652
SDP1	-0.18	-1.25	+626
SDP2	-0.28	-1.36	+662
SDP3	-0.29	-1.37	+618

Table 9. The vacuum faces sit behind a full assembly width of water, so the boundary treatment hardly matters here: Robin is 0.2–0.3 pcm below Marshak, and the builder's default zero-flux faces are 0.2–1.4 pcm below. The pin-area computation matters much more. With the default sampled fractions, SP1 is 226 pcm higher at $s = 4$ and 22 pcm lower at $s = 8$ than with exact fractions, the result of a fuel-area error of -0.66% and $+0.06\%$ (finding 4 explains why sampling errs this way). Pin-cell homogenization raises k by 379 pcm (SP1) to 662 pcm (SDP2) against the pin-resolved model at the same mesh.

4.5 Pin powers

Table 10 and Fig. 5. Unlike k, the pin powers improve with angular order: AVG falls from 1.43% for diffusion to 0.74% for SP3, 0.66% for SDP1 and 0.53–0.57% for SP5, SP7, SDP2 and SDP3. These are the paper's Table 7 values within 0.04 percentage points for SP1 to SP7, SDP1 and SDP2 (SP1 1.43 against 1.43, SP3 0.74 against 0.75), and the paper's RMSE measure agrees as well. The agreement shows that the paper scored its pin powers against the NEA MCNP reference used here, and that the two codes' solutions match pin by pin, not only in k.

For the higher orders AVG converges with mesh more slowly than k and from below: SP7 rises from 0.47% at $s = 8$ to 0.54% at $s = 48$, SDP3 from 0.49% at $s = 8$ to 0.56% at $s = 24$, still climbing. The remaining gap to the paper for SP7, SDP2 and SDP3 (0.03–0.08 points, largest for SDP3 at its coarser finest mesh) is consistent with that. Pin-power results from $s \leq 12$ therefore understate the converged error.

The largest pin error is 2.6–2.9% for every transport method and 5.8% for diffusion. In every case it is a corner pin of the outer UO_2 assembly next to the reflector, where the power is lowest (0.23–0.29) and the

Table 10: C5G7 pin powers against the NEA MCNP reference, Marshak vacuum, finest mesh. Errors in %; the paper's Table 7 AVG and RMSE in brackets. AVG, RMSE as the paper's tables use it ($\sqrt{\sum e_i^2}/N$), RMS, MRE and the largest pin error, the largest pin power (MCNP 2.498), and the assembly power errors.

Method	s	AVG (paper)	RMSE (paper)	RMS	MRE	max $\lvert e\rvert$	$P_{\max}$	inner UO_2	MOX	outer UO_2
SP1	48	1.43 (1.43)	0.053 (0.05)	1.72	1.22	5.79	2.522	+0.57	-0.45	-0.65
SP3	48	0.74 (0.75)	0.027 (0.03)	0.88	0.67	2.81	2.516	+0.46	-0.52	-0.06
SP5	48	0.57 (0.59)	0.021 (0.02)	0.69	0.51	2.93	2.500	+0.13	-0.16	+0.03
SP7	48	0.54 (0.57)	0.021 (0.02)	0.67	0.49	2.88	2.491	-0.10	+0.10	+0.05
SDP1	24	0.66 (0.68)	0.024 (0.03)	0.79	0.60	2.76	2.512	+0.38	-0.43	-0.02
SDP2	24	0.53 (0.57)	0.020 (0.02)	0.65	0.49	2.75	2.492	-0.06	+0.05	+0.03
SDP3	24	0.56 (0.64)	0.022 (0.03)	0.72	0.52	2.65	2.481	-0.34	+0.38	+0.06

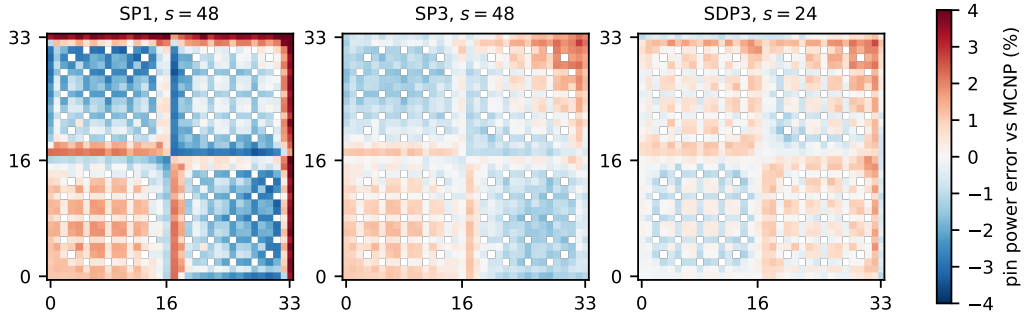


Figure 5: Pin power error against MCNP in the four fuel assemblies (inner UO_2 at the bottom left, MOX on the off-diagonal, outer UO_2 at the top right).

MCNP reference itself is least precise. The largest pin power is within 1% of MCNP's 2.498, and every assembly power within 0.65%.

4.6 Flux along the assembly diagonals

Fig. 6 and Table 11. `GARNET`'s SDP3 deviates from OpenMC by 1.1–1.7% of the peak (RMS) on the four panels, the paper's SDP3 by 1.1–2.0%: the two agree with the Monte Carlo profile equally well, and `GARNET`'s SDP3 is closer in group 7 of the MOX diagonal (1.7 against 2.0%). `GARNET`'s SP3 is further away, 1.5–1.9%, as in the paper's own SP_N/SDP_N comparison. The largest local deviations, up to 8% of the peak in group 7, sit on the thermal-flux peaks in the water between pins and at the reflector-side corner of the MOX assembly. The OpenMC curve is steepest there, so these are also the points where the extracted reference is least precise.

4.7 The homogenized model of the regression test

The regression test's model, pin-cell homogenized at $s = 1$, lands 8–165 pcm below MCNP (Table 12). The agreement is a cancellation of two large errors. Homogenizing each pin cell raises k by 380–660 pcm against the pin-resolved model (Table 9), so the homogenized mesh limit is 57–223 pcm *above* MCNP and 376–650 pcm above the paper; the $s = 1$ mesh then lowers k by 222–231 pcm. The test's eigenvalue envelopes (2% for diffusion, 0.5% for the family) are therefore not accuracy statements. What those tests guard is the internal ordering of the hierarchy and the clustering of the six transport methods, which is where the paper's SDP2/SDP3 table errors showed.

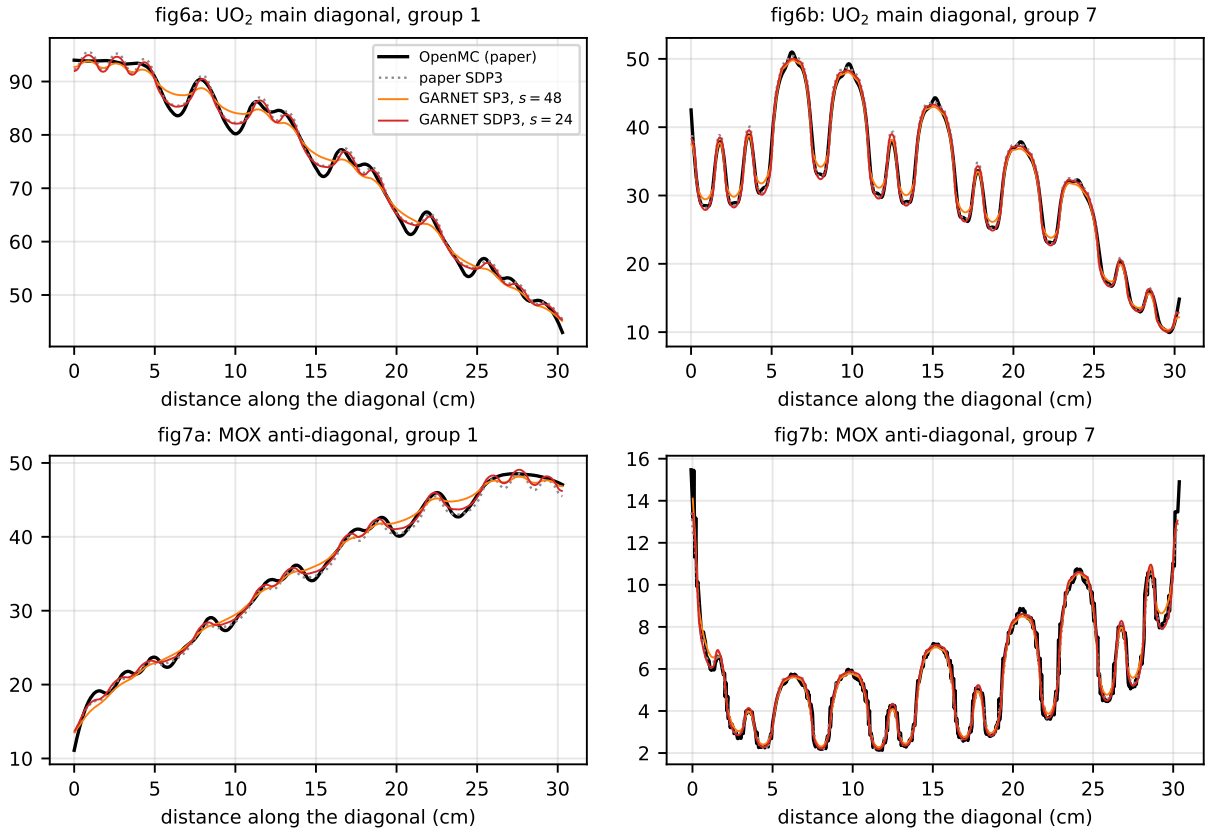


Figure 6: Groups 1 and 7 along the inner UO_2 main diagonal (paper Fig. 6) and the MOX anti-diagonal (Fig. 7): OpenMC and the paper's SDP3, both read from the paper, and `GARNET` SP3 and SDP3 scaled to OpenMC by least squares.

4.8 Solver behaviour and cost

Table 13. On one CPU thread at $s = 4$ (41 616 cells, seven groups) the SP_N solves take 6–40 s and the SDP_N solves 49–124 s. With Marshak vacuum every SDP_N block is non-symmetric and runs GMRES. At $s = 4$ its outer iteration count is still close to SP_N's (64–106 against 59–68), but from $s = 8$ on it roughly doubles (121–161 against 57–73), and at $s = 24$ the inner applies are 2.9–3.6 times those of SP3–SP7. On the GH200 at $s = 24$ (1.5 million cells) SP_N takes 30–271 s and SDP_N 1580–1969 s. At $s = 32$ none of the SDP_N solves converged: in the first outer iteration GMRES with Jacobi preconditioning stopped at 5000 iterations, 1.3–4.8 times above its tolerance (finding 5 in Section 6). The SP_N solves ran to $s = 48$ (6.0 million cells) in 2–14 minutes each.

The pin-resolved study is a GPU calculation: on the shared CPU node, with up to 16 runs at once, the SP_N solves at $s = 12$ took 14–77 minutes each and an SDP3 solve at $s = 12$ was projected at over 20 hours.

Table 11: Flux along the diagonals against OpenMC, after a least-squares scale (% of the OpenMC peak).

Figure	Solution	RMS (% of peak)	max (% of peak)
fig6a	paper SDP3	1.10	2.49
fig6a	GARNET SP3	1.72	4.18
fig6a	GARNET SDP3	1.08	2.32
fig6b	paper SDP3	1.36	8.15
fig6b	GARNET SP3	1.49	8.49
fig6b	GARNET SDP3	1.21	5.64
fig7a	paper SDP3	1.22	5.23
fig7a	GARNET SP3	1.86	4.66
fig7a	GARNET SDP3	1.22	4.72
fig7b	paper SDP3	2.04	15.22
fig7b	GARNET SP3	1.90	10.86
fig7b	GARNET SDP3	1.70	8.31

Table 12: C5G7 pin-cell homogenized, zero-flux outer faces. k_∞ from $s = 4, 8$ at second order. Differences in pcm.

Method	$s = 1$ (test)	$s = 2$	$s = 4$	$s = 8$	k_∞	test − MCNP	k_∞− MCNP	k_∞− paper
SP1	1.18490	1.18650	1.18696	1.18708	1.18712	-165	+57	+376
SP3	1.18611	1.18779	1.18830	1.18846	1.18851	-44	+196	+580
SP5	1.18630	1.18795	1.18843	1.18858	1.18863	-25	+208	+642
SP7	1.18640	1.18806	1.18852	1.18866	1.18870	-15	+215	+639
SDP1	1.18617	1.18784	1.18834	1.18850	1.18855	-38	+200	+613
SDP2	1.18639	1.18805	1.18851	1.18865	1.18869	-16	+214	+650
SDP3	1.18647	1.18816	1.18861	1.18873	1.18878	-8	+223	+592

Table 13: C5G7 cost, Marshak vacuum. CPU: best of three at $s = 4$, one thread. GH200: single runs, which include CuPy kernel compilation on first use; outer and inner counts at $s = 24$.

Method	Krylov	outers	inners	CPU $s = 4$ (s)	GH200 $s = 8$ (s)	outers $s = 24$	inners $s = 24$	GH200 $s = 24$ (s)
SP1	CG	61	17347	6.0	9.2	60	129535	30.5
SP3	CG	62	18641	15.4	61.8	73	161131	245.1
SP5	CG	68	20778	29.1	69.8	64	141625	232.5
SP7	CG	59	17579	40.0	76.7	67	143091	270.6
SDP1	GMRES	64	17242	49.0	223.8	144	509771	1590.2
SDP2	GMRES	90	20432	84.2	157.6	141	462018	1579.7
SDP3	GMRES	106	22254	124.3	191.9	158	464549	1968.7

5 Test tolerances

`tests/validation/test_brantley_larsen_2d.py`. The single-mesh tests now check the Marshak results, the default, on the 80×80 mesh: each method within 150 pcm of Table 6 (measured +87 to +127 pcm, all mesh error) and a positive offset with a spread below 60 pcm (measured 41 pcm). A new test keeps the per-moment Robin results for comparison: Robin minus Marshak between −80 and −45 pcm (measured −53 to −54 pcm, and −73 pcm for SDP1). Before, the single-mesh tests ran each solver's default, mostly Robin, and passed partly by cancellation (Table 3). The mesh-limit test was the weak one: it checked SP3 and SDP1 only, to 10 pcm. It now checks every method with the Marshak vacuum against Table 6 to 4 pcm, from $n = 80$ and 160 at second order; that two-mesh estimate is within 2.2 pcm of the three-mesh limits of Table 1, and the measured distances from the paper are within 1.5 pcm. SP1, SP3, SP5, SP7 and SDP1 run by default (about 20 s together); SDP2 and SDP3 with the FEMFFUSION tables, which need GMRES, are marked `slow`, as is a test that the re-derived SDP3 stays 4–12 pcm below the paper (measured −7.9 pcm from two meshes). A value near zero there would mean the tables had been changed back to the paper's.

`tests/validation/test_c5g7.py`. The homogenized tests keep their envelopes, with corrected explanations: Section 4 shows that their closeness to MCNP is a cancellation, so they guard the ordering and clustering of the hierarchy, not accuracy. Two tests are new. One checks that the exact pin areas reproduce πr^2 at any resolution and that the builder passes them and the vacuum boundary through. The other, marked `slow` (about a minute), runs pin-resolved diffusion at $s = 8$ with exact areas and vacuum outside, and checks it against the paper's SP1: k within 10 pcm (measured −6.8 pcm) and the pin-power AVG against the NEA MCNP map within 0.06 percentage points of the paper's 1.43% (measured 1.46%). It skips without a FEMFFUSION checkout, which carries that map.

6 Findings and recommendations

1. **Code verification passes.** On Brantley–Larsen, with the paper's Marshak boundary, `GARNET`'s mesh limits equal Table 6 within 0.6 pcm for SP1, SP3, SP5, SP7, SDP1 and SDP2, and within 0.2 pcm for SDP3 run with the authors' FEMFFUSION tables. On C5G7 the pin powers reproduce the paper's AVG and RMSE within 0.04 percentage points.

2. **The paper's SDP3 results use tables that differ from both the printed Eq. (A.6) and the derivation.** The FEMFFUSION tables reproduce the paper's SDP3 eigenvalue; the printed ones are 24 pcm off, and `GARNET`'s re-derived ones 10 pcm (Brantley–Larsen) to 14 pcm (C5G7) lower. `GARNET` keeps the re-derived tables. No code change is needed; the difference is documented in `garnet/spn.py`.

3. **The vacuum default was inconsistent across the family; Marshak is now the default.** When this study started, only the dedicated SP3 solver defaulted to the coupled Marshak condition. SP5, SP7 and the SDP$_N$ solvers defaulted to the per-moment Robin term, which is 52–73 pcm low on Brantley–Larsen and depends on the moment variables it is applied to (0.35–0.7 pcm between equivalent solvers); on a reflected core such as C5G7 the choice is irrelevant (0.3 pcm). Following this study, every SP$_N$/SDP$_N$ eigen, transient and noise solver applies Marshak by default wherever the coupled term exists (a Cartesian grid with a vacuum face and no hybrid mask) and Robin elsewhere; `marshak_vacuum=False` restores Robin for comparison. This moves SP5, SP7, SDP2 and SDP3 results on problems with vacuum faces, by about +53 pcm on Brantley–Larsen. SDP2 pays for it: it loses its CG path and runs GMRES, 5.8 times slower at $n = 160$ (Table 7); SDP3 already ran GMRES, and SP$_N$ keeps CG. The dedicated SDP1 solver, eigen and transient, keeps Robin, because the coupled matrix is not symmetric in its two-moment basis; the general SDP$_N$ block of order 1 gives SDP1 with Marshak. Very large SDP$_N$ meshes are subject to finding 5.

4. **The C5G7 builder's sampled pin areas are biased.** In the pin-resolved model, each cell that the $r = 0.54$ cm fuel circle cuts is a blend of pin and water weighted by its covered fraction, so a pin's fuel inventory is the sum of those fractions. By default each fraction is estimated by counting which of 16×16 sample points in the cell fall inside the circle. All 1056 fuel pins are cut identically, so this error does not average out over the core: it is a systematic error in the fuel inventory, of irregular sign and size, −0.66% at $s = 4$, +0.12% at 6, +0.06% at 8 and −0.02% at 16. In this lattice k moves

by about −35 pcm per +0.1% of fuel area, so the sampled model is 226 pcm high at $s = 4$ and 22 pcm low at $s = 8$, more than any difference between methods in this study and enough to make a mesh study irregular. With `exact_area=True` each fraction is the closed-form area of the square–disk intersection (the routine of the HP-MR fix, audit item O6), and the fuel inventory equals πr^2 per pin to round-off on every mesh. *Recommendation:* make exact areas the default. No test or stored result depends on the sampled default (the colorset ADF study already samples 256×256 points); it would change the output of the examples `c5g7_meshconv.py`, `sdpn_c5g7.py --pin-resolved` and `c5g7_td.py --pin-resolved`, whose C5G7-TD builder uses the same sampling. The builder's zero-flux outer faces are within 1.4 pcm of vacuum and can stay.

5. **The non-symmetric SDP$_N$ solve does not scale to the finest C5G7 meshes.** With Marshak vacuum, SDP1, SDP2 and SDP3 at $s = 32$ (2.7 million cells) all stopped in the first outer iteration: GMRES with the default Jacobi preconditioner did not reach its tolerance in 5000 iterations (residual 1.3–4.8 times the target). Up to $s = 24$ they converged, but with 121–161 outer iterations against 57–73 for the CG-solved SP$_N$ blocks. Tracked as audit item O14. *Recommendation:* precondition the large non-symmetric blocks (for example `spn_precondition`, already available) and decide whether an inner solve that misses its tolerance in an early outer iteration should abort the eigenvalue solve.

6. **Mesh requirements.** Brantley–Larsen on the 80×80 test mesh carries +87 to +132 pcm of mesh error. Pin-resolved C5G7 needs at least 16 cells per pin side for a 4 pcm eigenvalue; at 4 cells it is 68–216 pcm off, and its convergence is not monotone.

7. **C5G7 offset.** `GARNET`'s pin-resolved C5G7 eigenvalues are 11–22 pcm below the paper's, growing with angular order. The models differ in the pin-boundary representation, which the paper does not specify; the offset is 3–5% of the angular error and is left unexplained.

8. **The C5G7 regression test's agreement with MCNP is a cancellation** of +380 to +660 pcm homogenization error and about −225 pcm mesh error. Its docstrings now say so; its envelopes are kept.

9. **Devices agree.** CPU and GH200 give the same k to 1.3×10^{-13} in 18 configurations.

A Reproduction

From the repository root, with `GARNET` importable (`PYTHONPATH=.`):

```
JOBS=8 bash benchmark-results/carreno2024-sdpn-vv/run_sweep.sh all   # CPU runs
bash slurm/stage_and_submit_carreno_gh.sh                            # GH200 runs
GARNET_VV_STAGE_ROOT=<new dir> GARNET_VV_ROBIN_CELLS= \
  GARNET_VV_CELLS="4 6 12 20 32 48" GARNET_VV_METHODS="SP1 SP3 SP5 SP7" \
  bash slurm/stage_and_submit_carreno_gh.sh                          # GH200 batch 2
#   (and GARNET_VV_CELLS="4 6 12 20 32" GARNET_VV_METHODS="SDP1 SDP2 SDP3")
bash benchmark-results/carreno2024-sdpn-vv/run_timing.sh             # timings
cd benchmark-results/carreno2024-sdpn-vv && make
```

`tools/assess_carreno2024.py` writes one JSON record per run: k, convergence, outer and inner iterations, setup and solve seconds, the flux on the paper's plotted lines and, for C5G7, the 51×51 pin powers. `data/runs/` holds the CPU records (merlin7 `cn000`, one thread per run, up to 16 runs at a time; `progress.log` lists them, including the runs stopped as too slow), `data/runs-gh200/` the GH200 records with their job logs and staging provenance (two batches, both commit `20967d3` plus the uncommitted audit work), and `data/timing/` the best-of-3 timings, taken with nothing else of this study running. The SDP3 table variants (`SDP3-A6`, `SDP3-FFN`) were run with the same tool outside the sweep. `data/paper-figures.json` is the output of `extract_paper_figures.py` (it needs PyMuPDF; the report build does not). `data/nea-c5g7-mcnp-pin-powers.json` is the NEA MCNP pin-power map as distributed with FEMFFUSION (`examples/2D_C5G7/c5g7_transport.ref`, SHA-256 recorded in the file); its assembly powers are 492.80, 211.71 and 139.78 and its largest pin power 2.498, the NEA values. `make_figures.py` turns all of these into `figures/`, `tables/` and `data/summary.json`.